

# SparkADS

SparkADS is a Beckhoff TwinCAT 2/3 ADS/AMS driver that talks ADS directly to the PLC. No TF6100 OPC UA server license is required.

Version 0.1.1 Ignition 8.1 and 8.3 Runs on: any gateway that reaches the PLC network

## Contents

1. [1. Overview](#)
2. [2. What You Need](#)
3. [3. Install](#)
4. [4. License](#)
5. [5. Configure](#)
6. [6. Operate](#)
7. [7. Verify](#)
8. [8. Troubleshoot](#)
9. [9. Reference](#)
10. [10. Support](#)

## 1. Overview

SparkADS is a standalone device driver, licensed and running independently of any other SparkBridge module. Module id `com.greenguardusa.sparkads`.

## 2. What You Need

- Ignition 8.1 and 8.3.
- A TwinCAT 2 or 3 PLC reachable over TCP 48898 (ADS/AMS), with a static AMS route configured on the PLC pointing to the gateway's AmsNetId.

## 3. Install

Install the signed module on Ignition 8.1 or 8.3 through Config > Modules > Install or Upgrade a Module; no build step is required. Module id `com.greenguardusa.sparkads`.

## 4. License

Licensed per gateway, bought once and kept, soft-licensed (`com.greenguardusa.license`, entitlement ADS).

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

## 5. Configure

Config > OPC UA > Device Connections > Create new Device > the ADS device type.

|                                  |                                                                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Setting</b>                   | Meaning, default                                                                                                                                           |
| <b>Hostname/IP, Port</b>         | PLC address; AMS/TCP port 48898.                                                                                                                           |
| <b>Target NetId, Target Port</b> | The PLC's AmsNetId and AMS port. TwinCAT 3 PLC1 defaults to port 851 (801/811/821/831 are TwinCAT 2 runtimes 1-4; 851/852/853 are TwinCAT 3 runtimes 1-3). |
| <b>Local NetId</b>               | This driver's own AmsNetId, which must be added as a route on the PLC.                                                                                     |

Manual tag addressing: `MAIN.x` (symbolic), `%MW100:INT`, or `0x4020,0x100:DINT` (index group/offset).

## 6. Operate

The PLC must have a static AMS route pointing to the gateway's Local NetId before the device can connect; a device that never leaves DISCONNECTED usually means the route is missing or the Target NetId/Port do not match the PLC's runtime.

## 7. Verify

See the README in the bundle for the verification procedure.

## 8. Troubleshoot

|                                   |                                                                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symptom</b>                    | Cause / fix                                                                                                                                                                   |
| <b>Device will not connect</b>    | Confirm the AMS route on the PLC points to the gateway's Local NetId, and that Target NetId/Target Port match the PLC's actual runtime (TwinCAT 2 vs 3, PLC instance number). |
| <b>Symbolic address not found</b> | Confirm the symbol name matches the PLC project exactly, including case, for example <code>MAIN.x</code> .                                                                    |

## 9. Reference

|                                    |                                                |
|------------------------------------|------------------------------------------------|
| <b>Module id</b>                   | <code>com.greenguardusa.sparkads</code>        |
| <b>Default port</b>                | TCP 48898 (ADS/AMS)                            |
| <b>TwinCAT 3 PLC1 default port</b> | 851                                            |
| <b>License entitlement</b>         | ADS ( <code>com.greenguardusa.license</code> ) |

## 10. Support

Docs page section: </sparkbridge/docs#sparkads>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).